

Clinical Risk Assessment Report: Patient 52

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 45.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.07x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +6.8%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT)	Value: -0.42 [Bottom 2%] (+6.8%)
<i>Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.</i>	
Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT)	Value: 1.04 [Top 13%] (+4.0%)
<i>Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.</i>	
Small Zone Emphasis (PET) (GLZLM_SZE.PET)	Value: 0.15 [Avg (55th %ile)] (+3.3%)
<i>Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.</i>	
Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)	Value: -0.23 [Bottom 9%] (+2.5%)
<i>Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.</i>	
Maximum Tumor Density (CONVENTIONAL_HUmax)	Value: 0.53 [Avg (75th %ile)] (+2.5%)
<i>Reflects the most dense solid components or calcifications within the primary tumor lesion.</i>	
75th Percentile Tumor Density (CONVENTIONAL_HUQ3)	Value: 0.75 [Avg (73th %ile)] (+2.3%)
<i>Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.</i>	
Squamous Cell Carcinoma Antigen (SCC_Ag)	Value: 0.79 [Avg (38th %ile)] (+2.0%)
<i>Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.</i>	
High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT)	Value: 0.99 [Top 16%] (+2.0%)
<i>Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.</i>	

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

